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Secured Reversible matrix embedding based on dual image using Integer wavelet and Arnold Transform

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Abstract

Steganography is the art and science of hiding information. In Steganography host image is destroyed and cannot be attained to the original state. In a sensitive area such as military and medical recovery of the original cover medium is required without any loss. A special case of data hiding that retrieves the cover media and secret information without a loss of a pixel is known as reversible data hiding. In this paper reversible data hiding technique is proposed to improve the security and enhance the capacity; a hybrid model of integer wavelet transform and Arnold transform are used. The cover image is decomposed into the integer wavelet sub-band; each sub-band is further scrambled using Arnold transform before embedding scrambled secret information. In this work dual image embedding mechanism is used such that it makes matrix embedding reversible without any bookkeeping data. The experimental result shows that the proposed method provides high embedding capacity.

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1. Introduction

Embedding secret information in different medium such as image, video, audio is called Data Hiding.

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There is a tradeoff between the capacity of information stored to the quality or distortion of the image after embedding. The quality of image is directly proportion to payload and embedding technique. Thus

Reversible data hiding (RDH) is a special case which came into existence where at the receiver end, after retrieval of secret information the original medium is intact. This concept is mainly beneficial for military, medical applications and remote sensing.

RDH can be carried out into spatial and frequency domain [1]. Spatial domain directly deals with intensity of the pixels and in frequency domain the coefficient of pixels are modified to embed the information.

In recent years many RDH techniques have been proposed, these can be categorized broadly as compression based, histogram shifting based, difference expansion and frequency based techniques: which comprises of Discrete Cosine Transform (DCT), Discrete Fourier Transform (DFT), Discrete Wavelet Transform (DWT) and integer Wavelet Transform (IWT).

2. Preliminaries

Integer wavelet Transform and Arnold Wavelet Transform are discussed in the section below

2.1. Integer Wavelet Transform

2D Integer wavelet Transform is one of the major reversible data hiding technique. Spatial image is converted to frequency domain using lifting steps. The lifting scheme [1] [2] consists of three steps:

The basic Haar wavelet is defined as

$$Average = (Odd + Even) / 2 \quad (1)$$

$$Difference = (Even - Odd) \quad (2)$$

Split

The wavelet first splits the signal into odd and even samples.

Predict (Dual Lifting)

In this step a filter is applied to even samples and subtracted it from the odd samples.

$$NewOddSet = (EvenSet - OddSet) \quad (3)$$

Update (Primary Lifting)

In this the average of odd samples and the difference computed using Equation 3 is used as below

$$NewEvenSet = \left(Oddset + Floor \left(\frac{NewOddSet}{2} \right) \right) \quad (4)$$

The inverse transform is applied by reversing the operation and flipping the signs to get the original information.

Inverse Primary Lifting: The predicted and updated set is subtracted to get back the original even set

$$OriginalEvenSet = \left(NewEvenSet - ceil \left(\frac{NewOddSet}{2} \right) \right) \quad (5)$$

Inverse (Dual Lifting): The addition is performed on the update step result and the predict step result

$$OriginalOddSet = (OriginalEvenset + NewOddset) \quad (6)$$

Merge: This Original Odd and even sets are merged to get back the sequence of data.

2.2. Arnold Transform

Arnold cat map or Arnold transform was first presented by Vladimir Arnold in 1960 [3]. This is a chaotic map and it scrambles the image so it becomes noisy and imperceptible.

The equation for Arnold Transform is as below

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{pmatrix} x & + & y \\ x & + & 2y \end{pmatrix} (\text{mod } N) \quad (7)$$

Where $x, y \in (0, 1, 2, \dots, N-1)$ where N is the order of image and x' and y' are new coordinates of the pixels. For each N after K iterations the original image is obtained. For example for an image of order 256×256 if the Arnold map is scrambled for 192 times, original Image is obtained.

3. Related Work

There are many contributions in the field of steganography to achieve Non-repudiation, concealed information, and proof of ownership, etc. In early days concealment of information was achieved by directly modifying the intensity values of pixels in the spatial domain which has drawbacks of distortion and information was easily traceable. To overcome these drawback various studies found that transforming the image in the frequency domain and using the coefficients to hide the information makes the data invisible and more secured than spatial image.

The work in the literature [4] presented a multi-image technique that uses Arnold Transform to hide three 8-bit secret images within a single color cover image of 24 bits. The secret information is encrypted and the 3 MSB bits of each secret image are embedded in the last 3 LSB bits of red, green and blue pixels. The distortion is agreeable and there is a scope to improve the capacity and histogram error between stego images and cover image needs improvement.

A random diffusion and Arnold transform-based scheme [5] is proposed to increase security. The secret message bits are embedded using LSB algorithm and Arnold transform scrambles the pixels and further diffusion is done using a pseudo-random generator. The payload capacity is low using this method.

In literature [6] first the secret image has been scrambled using Arnold algorithm and embedded into the cover image by using LSB algorithm. It is used for blind steganalysis of bmp images. The main features of the proposed method are imperceptibility and security.

A technique where combination of DWT, DCT and SVD is applied with training the system using back propagation neural network (BPNN) is presented [7]. DCT is applied on third level DWT sub-band and SVD is applied on the obtained DCT coefficient. The watermarking information is further scrambled using Arnold transform and DCT; DWT is performed on the resultant image. Then embedding is performed into the SVD-based DCT coefficient of cover image

A histogram modification based system has been applied [8] to prevent underflow/overflow problem. Here cover image is divided into 8×8 block and converted to frequency domain using IWT. The hiding capacity of each coefficient in the sub-band is calculated based on modified range to increase the capacity. Basically each coefficient embeds different number of bits. The coefficients after embedding remains in the range are considered. The pseudo

random generator is used to select the embedding locations to improve the security [9]. Optimal pixel adjustment (OPA) algorithm is used to minimize the error difference between original and modified coefficients.

A JPEG-based compression and watermarking technique is presented. The study [10] [11] combines the watermarking and compression technique to reduce computation cost and time. In this method image is divided into 8×8 blocks and DCT is applied on each block and water mark is embedded and image is compressed after quantization. At the receiver end each block is dequantised and inverse DCT is applied on the extracted watermark and finally decompressed to retrieve original watermark.

A novel reversible data hiding scheme [12] is proposed using integer wavelet Transform, histogram shifting and orthogonal decomposition. Histogram shifting is carried out in the high frequency coefficients as it has Laplacian distribution. The reversibility of the data hiding operation is facilitated by the orthogonal coefficients. In this work first IWT is performed on the cover image and all the coefficients in the frequency domain are encrypted; information is then embedded in the encrypted coefficients. To obtain the stego image inverse IWT is applied and image is transmitted.

As per the literature [13][13][15] it is found that majority methods use LSB substitution and Matrix encoding[16] can be used for better embedding technique. The matrix encoding is not a reversible technique.

This paper uses [17] wavelet transform and Arnold transform. Secret data is scrambled before embedding into the cover image. Secret data is embedded into significant wavelet coefficients of the DCT sub bands of a cover image.

From the above literature we have found that that capacity can be increased and reversibility should not be limited to only watermarking. There is a scope of reversible data hiding where there is a tradeoff between capacity and imperceptibility and matrix encoding [15] can be enhanced to obtain reversibility which has not been attempted.

4. Proposed System

In the proposed system the cover image is first decomposed to high frequency and low frequency sub bands using Integer Wavelet Transform (IWT). For enhancing the security the decomposed sub bands are scrambled using the Arnold transform and even the secret image is scrambled before embedding in the cover image. The process of embedding and extraction are showed in Figure1 and Figure 2

To embed the information we use two techniques

1. 2 bit LSB Embedding
2. Matrix Embedding.

To increase the capacity, improve the security and obtain the cover image lossless, once secret image is retrieved; we generate two copies of the cover image and hide information in different bit positions.

For example: In 2 bit LSB embedding using a gray scale 8 bit cover image 6th and 7th bit are utilized for hiding the information then in 2nd copy of cover image, 4th and 5th bits are used. The wavelet sub bands coefficients are real numbers with negative values we cannot use those coefficients. In order to make use of all coefficients we take 2's complement and represent each value as a 16 bit and perform embedding to retrieve each pixel without any loss. Once extraction process is completed the final copy is represent as 8 bit gray scale image. This enables us to utilize more bits for hiding.

The embedding and extraction are shown in Figure 1 and 2 respectively.

4.1. Embedding

- Step1: Apply IWT on the cover image to obtain low and high frequency sub band. The low frequency sub-band is further decomposed to 2nd level
- Step2: Apply Arnold Transform on secret image and convert resultant to binary pixels
- Step 3: Apply Arnold Transform to scramble the positions in all sub band except 2nd level LL sub band
- Step 4: Make two copies of the each sub band
- Step 5: Embed secret image in using 2 bit Random LSB in scrambled sub band (first two bits in copy 1 and next two bits in copy 2) or using matrix embedding (last 3 bits to hide two bits) (4,5,6 to hide next two bits).
- Step 6: Unscramble and apply IWT to obtain stego image in spatial domain

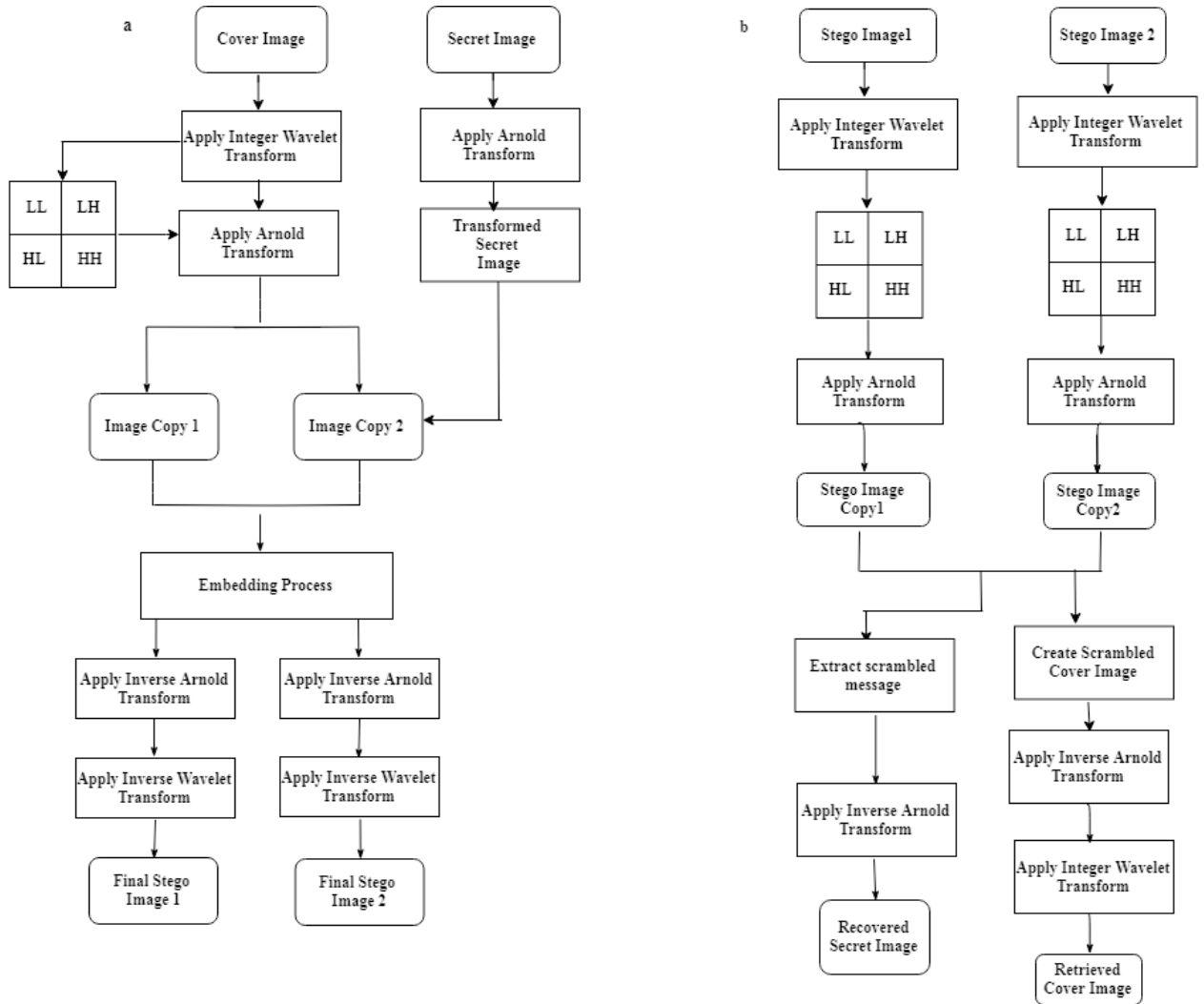


Figure1. (a)Embedding process (b) Extraction process

4.2. Extraction

- Step 1: Apply IWT on the stego image copy 1 and copy2
- Step 2: Apply Arnold Transform to obtain the scrambled sub bands
- Step 3: Retrieve scrambled secret image
- Step 4: Apply inverse Arnold transform to obtain secret image.
- Step 4: Rebuild the cover image using copy 1 and copy2
- Step 5: Apply inverse Arnold Transform and IWT to retrieve original cover image

5. Results and Discussion

For experiment purpose we have used 8-bit grayscale images of size 512×512 for cover images. We have used three grayscale 8-bit fingerprint secret images of size 350×350 for testing purpose. The standard test images used are Lena, Baboon and Jet. Integer-to-integer wavelet transform is applied on each image and decomposed into high and low frequency sub-bands. The low frequency sub band is further decomposed into 2nd level. The sub band used for hiding are (LH, HL, HH and LH2, HL2, HH2). We use PSNR to calculate the distortion between stego image and original image. Higher the PSNR denotes lesser the distortion and better quality of image. If the PSNR is lesser than 30dB, then distortion is more visible. Given below is the equation to calculate PSNR and in turn MSE.

$$PSNR = 10 \frac{\log_{10} (max)^2}{MSE} \quad (8)$$

Here max is maximum pixel value of an image.

$$MSE = \frac{1}{MN} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i,j)^2 - I_s(i,j)^2] \quad (9)$$

Where I is the cover image and I_s is the stego image.

Structural similarity index SSIM is used for comparing the similarity between two images and is calculated as given in the below equation

$$SSIM(x, y) = \frac{(2\mu_x \mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 \mu_y^2 + c_1)(\sigma_x^2 \sigma_y^2 + c_2)} \quad (10)$$

Where μ_x is average of x_i

μ_y The average of y_i

σ_x^2 is variance of x_i

σ_y^2 is variance of y_i

The embedding rate is defined as the rate at which secret message can be hidden in the cover image and is calculated as given below:

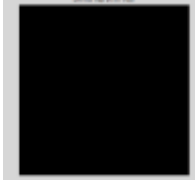
$$Rate\ of\ Embedding = \frac{Hidden\ bits}{No\ of\ pixels\ of\ cover\ image} \quad (11)$$

Table 1: Results of original cover image, Stego image 1 and 2, recovered cover image and the difference

Test Image	Cover Image	Stego Image1	Stego Image2	Retrieved cover image	Difference
(a)Lena		 PSNR= 43.1768	 PSNR =31.0299	 PSNR= $-\infty$, SSIM=1	

As shown in Table 1, we have two copies of stego image in which the test secret image of size 350x350 has been embedded. On retrieval without any attack cover and secret image are recovered completely without loss and that is reconfirmed by PSNR of the recovered image with original cover and secret image which is ∞ . Results of recovered original image, secret image and their difference are shown in Table2.

Table 2 Results of original secret image, recovered secret image and the difference

Secret Image	Cover Image	Retrieved cover image	Difference
Fingerprint			

The proposed framework has been tested with standard images like Lena, Airplane and Mandrill. As compared to literature [18] the embedding capacity and PSNR is higher and all the coefficients are used to hide as compared to only zeros in [19] and [20]. This methodology do not have any adhoc of indexing table except for reserved space for the size of the image and iteration number of Arnold transform in the last two rows. Table 3 shows the embedding capacity, PSNR, SSIM and bpp.

Table 3: Embedding capacity, PSNR, SSIM and bpp of proposed and existing system

Embedding Capacity			PSNR (dB)		SSIM		Embedding Rate (bpp)		
Test Images	Existing method[8]	Proposed method	Existing method	Proposed method		Proposed method		Existing method	Proposed method
				Copy 1	Copy 2	Copy 1	Copy 2		
Lena	310520	980000	33.56	43.1768	31.0299	0.9585	0.5669	1.185	1.86
Airplane	309216	980000	31.07	43.1032	30.4271	0.9604	0.5600	1.180	1.86
Mandrill	308032	980000	23.89	43.5459	31.0296	0.9792	0.6818	1.175	1.86

6. Conclusion and Future Enhancement

This paper proposes a reversible data hiding method based on integer-to-integer wavelet transform and Arnold Transform. The dual image embedding does not require any additional memory of booking data which in turn enhances the embedding capacity. This work has achieved in reversing the matrix embedding successfully. Experimental results show higher embedding capacity as compared to existing work. In Future the work can be further enhanced to improve the parameters and use machine learning techniques to achieve better results and train the images such that it can withstand attacks and retrieve the cover and secret image completely by fixing any tampering caused while transmission.

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